The due date for submitting this assignment has passed.
Due on 2026-04-10, 23:59 IST.

You may submit any number of times before the due date. The final submission will be considered for grading.

You have last submitted on: 2026-04-10, 20:31 IST

Let A be a $m \times n$ non-zero real matrix, where m and n are two non-zero natural numbers strictly greater than 1. Answer the following questions.

1) Choose the correct options from the following. ***2 points***

- [x] If $m < n$, then the maximum possible value of rank of A is m.
- [ ] If $m < n$, then the minimum possible value of nullity of A is n.
- [ ] If $m > n$, then the maximum possible value of rank of A is m.
- [x] If $m > n$, then the minimum possible value of nullity of A is 0.

Yes, the answer is correct.
Score: 2

Accepted Answers:

If $m < n$, then the maximum possible value of rank of A is m.
If $m > n$, then the minimum possible value of nullity of A is 0.

2) If A is the matrix representation of a linear transformation T from V to W, then choose the set of correct options. ***2 points***

- [ ] $dim(V) = m$.
- [x] T cannot be zero transformation.
- [x] If $m < n$, then T cannot be one to one.
- [ ] If $m < n$, then T cannot be onto.
- [x] If $m = n$, then T cannot be an isomorphism.

Yes, the answer is correct.

Score: 2

Accepted Answers:

T cannot be zero transformation.
If $m < n$, then T cannot be one to one.
If $m = n$, then T cannot be an isomorphism.

3) Let A_1 and A_2 be two affine spaces defined by the solution space of the system of linear equations $\begin{bmatrix}1 & 0 & 1\\ 0 & 1 & 1\\ 1 & -1 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} = \begin{bmatrix}2\\ 2\\ 0\end{bmatrix}$ and $\begin{bmatrix}1 & 0 & 1\\ 2 & 0 & 2\\ 3 & 0 & 3\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} = \begin{bmatrix}2\\ 4\\ 6\end{bmatrix}$ respectively. Let V_1 and V_2 be the vector subspaces of R^3 corresponding to A_1 and A_2. ***3 points***

Choose the correct options.

- V_1 has dimension 1, where as the dimension of V_2 is 2.
- V_1 has dimension 2, where as the dimension of V_2 is 1.
- Both the V_1 and V_2 have dimension 3.
- Both the V_1 and V_2 have dimension 1.

Yes, the answer is correct.

Score: 3

Accepted Answers:

V_1 has dimension 1, where as the dimension of V_2 is 2.

4) Consider the following subsets of R^3, with $\langle\ ,\ \rangle$ used to denote the standard dot product on R^3, and $\|\ \|$ used to denote the standard length function.

$$S_1 = \{v \in \mathrm{R}^3 \mid \langle v, (1,0,0)\rangle = 2,\ \|v\| = 2\}$$
$$S_2 = \{u \in \mathrm{R}^3 \mid u \in \mathrm{Span}\{(1,0,0)\},\ \|u\| = 2\}$$

If n denotes the number of elements in $S_1 \cap S_2$, then find the value of n.

1

Yes, the answer is correct.

Score: 3

Accepted Answers:

(Type: Numeric) 1

3 points

Consider a linear transformation $S : V \rightarrow W$ defined by $S(A) = AB$, where $B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$, V is the vector space of all upper triangular matrices of order 2 with respect to the usual addition and scalar multiplication in matrices, and W is the vector space of all matrices of order 2 with respect to the usual addition and scalar multiplication in matrices. Let $\beta = \{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\}$ be an ordered basis of V and $\gamma = \{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\}$ be an ordered basis of W and C be the matrix representation of S with respect to the ordered basis β for the domain and γ for the codomain. Use this information to answer the following sub questions.

5) Which of the following option(s) is/are true? ***2 points***

- [ ] Order of the matrix C is 3×4.
- [x] Order of the matrix C is 4×3.
- [x] C has no zero row.
- [ ] C has only one non-zero column.

Yes, the answer is correct.

Score: 2

Accepted Answers:

Order of the matrix C is 4×3.
C has no zero row.

6) Find the rank of S.

3

Yes, the answer is correct.

Score: 2

Accepted Answers:

(Type: Numeric) 3

2 points

7) Find the nullity of the matrix C.
0

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 0

1 point

8) A norm on a vector space V is a function ***3 points***

$$\begin{aligned} \|\cdot\| : V &\rightarrow \mathrm{R} \\ v &\rightarrow \|v\| \end{aligned}$$

satisfying the following conditions:

i) $\|v_1 + v_2\| \leq \|v_1\| + \|v_2\|$ for all $v_1, v_2 \in V$.
ii) $\|cv\| = |c|\|v\|$ for all $c \in \mathrm{R}$ and for all $v \in V$.
iii) $\|v\| \geq 0$ for all $v \in V$; $\|v\|$=0 if and only if $v = 0$.
Define $V = \mathrm{R}^2$ and the function defined as:

$$\begin{aligned} \|\cdot\| : V &\rightarrow \mathrm{R} \\ \|(x, y)\| &= max\{|x|, |y|\} \end{aligned}$$

Which of the conditions of a norm are satisfied for the above function?

- [x] Condition 1.
- [x] Condition 2.
- [x] Condition 3.

Yes, the answer is correct.

Score: 3

Accepted Answers:

Condition 1.

Condition 2.
Condition 3.

Suppose two publication houses (publication house A and publication house B) have organized a sale of their books. Both of them publish three types of books: novels, poetry collections and collections of short stories. The selling price (in (hundreds) ₹) of these three types of books in publication houses A and B are given as follows:

	Novels	Poetry collections	Collections of short stories
Publication house A	2	3	8
Publication house B	4	9	16

Table: MQ2M2T1

The publication houses announced that in order to avail these special sale prices, customers have to buy equal number of novels, equal number of poetry collection, and equal number of collection of short stories from each of the publication houses (i.e., if a customer buys x number of novels, y number of poetry collections and z number of collection of short stories from Publication house A; then they have to buy exactly x number of novels, y number of poetry collections and z number of collection of short stories from Publication house B, to avail the benefit of the sale). So there is a map taking the tuple consisting of the number of books of each type bought (Novels, Poetry collections, Collection of short stories) to the prices paid by customers who availed the sale to each of the publication houses, which yields a linear transformation (T) from R^3 to R^2 (where the first and second co-ordinates of the image denotes the prices paid to publication house A and publication house B, respectively).

Answer the following questions using the above information.

9) If A is the matrix representation of T with respect to the basis $\{(1,0,0),(0,1,0),(0,0,1)\}$ for R^3 and to the basis $\{(1,0),(0,1)\}$ for R^2, then A is ***2 points***

- $\begin{bmatrix} 2 & 4 \\ 3 & 9 \\ 8 & 16 \end{bmatrix}$
- $\begin{bmatrix} 2 & 3 & 8 \\ 4 & 9 & 16 \end{bmatrix}$
- $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Yes, the answer is correct.

Score: 2

Accepted Answers:

$\begin{bmatrix} 2 & 3 & 8 \\ 4 & 9 & 16 \end{bmatrix}$

10) Find out the rank of T.

2

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 2

1 point

11) If $\{(l, m, n)\}$ is a basis of $ker(T)$, then find the value of m if $n = 1$.

0

Yes, the answer is correct.

Score: 2

Accepted Answers:

(Type: Numeric) 0

2 points

12) If A is the matrix representation of T with respect to the basis $\{(1, 0, 1), (0, 1, 1), (1, 1, 0)\}$ for R^3 and to the basis $\{(1, 0), (0, 1)\}$ for R^2, then find out the sum of the elements of the bottom most row of the matrix A.

58

Yes, the answer is correct.

Score: 2

Accepted Answers:

(Type: Numeric) 58

2 points